

NumPy in Agentic AI (Simple Explanation)

What is NumPy?

NumPy is a Python library for fast numerical computation using arrays.

Where is NumPy used in Agentic AI?

1. Convert text into embeddings (vectors).
2. Store embeddings as arrays.
3. Compare embeddings using vector math (dot product, distance).
4. Rank the most relevant documents.
5. Process images (pixel arrays).
6. Process audio (waveform arrays).
7. Prepare data for machine learning.

Typical Agent Flow

User Question -> Embedding -> Similarity Search -> Retrieve Documents -> LLM -> Answer

Role of NumPy

NumPy is mainly used during embedding storage and vector calculations. LLMs generate the final answer, while NumPy provides the mathematical foundation behind retrieval and data processing.

Important NumPy Functions

`np.array()`, `np.dot()`, `np.linalg.norm()`, `np.argsort()`, `np.mean()`, `np.sum()`, `np.reshape()`.

Key Point

Think of NumPy as the math engine behind many AI systems. It helps AI work with vectors, matrices, images, audio, and machine learning data efficiently.